

Timing Tells and Decision-Making in High-Stakes Poker Tournaments

Group 8

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Section 1: Introduction to the Data

Our group selected Poker to generate insights for tournament players, allowing them to increase their EV (expected value) while playing. We hypothesized that analyzing the time it takes players to make their previous decisions would help to predict their future actions. This idea is discussed in the poker community as “timing tells,” but it lacks data-based analysis to justify its effectiveness and is more of a pseudoscience than anything else. The effectiveness of timing tells is quite a controversial topic within the poker community, so we set out to see if the data would back up this piece of poker theory.

We collected our data through watching multiple different days of the 2025 WSOP \$10k buy in main event. We chose the 2025 WSOP \$10k Main Event across multiple days because it offers a large, highly competitive player pool with consistently high stakes, ensuring that the decisions made are serious and strategically meaningful. Sampling across multiple days also helps reduce biases from specific table dynamics, fatigue levels, or individual player tendencies, making our data more representative of general tournament play. Each row represents an entire hand with the conditions that it must have only two players post-flop, with no pre-flop all-in. We each analyzed 50 hands for a total of 200 unique observations, with each observation containing 53 variables. We then proceeded to generate an additional 8 variables of their perceived hand strength (4 for each player, for preflop, flop, turn, and river) for 61 total variables.

Our data was mainly focused on 2 players, p1 (player 1), and p2 (player 2), where p1 is the player that acts first postflop. Any variable that contained p1 or p2 showed that the variable belonged to the respective player. The data was then further divided into each street of the hand: preflop, flop, turn, and river (abbreviated to pf, f, t, and r, then placed in front of the variable to show which street they belonged to). Across each street, we analyzed a lot of the same variables, such as stack size, action taken, time taken, bets placed, and much more.

p1_name	p1_stacksize	p1_position	p1_cards	p1_strength	pf_p1_hs	blinds	chip_leader	last_agressor	pf_pot_size	community_cards	f_p1_action	f_p1_time	f_bets	winner
Wigg	54600	BB	Qh_9h	47	0.595	200/400	83900	2	2600	9d_7c_Kd_4c_10c	ch-ca	2-9	900	1
Bleznick	72200	SB	3s_3c	8	0.543	200/400	161800	2	10800	4h_9h_6s_X_X	ch-r	12-43	4000-16500	1
Locquet	75800	BB	9d_4d	63	0.377	200/400	155300	2	2600	7h_Js_Kh_X_X	ch-f	3-2	1300	2
Afriat	82900	BB	Ac_3c	25	0.563	200/400	117300	2	2600	10s_6s_5h_X_X	ch-f	4-1	1200	2
Barnes	164600	SB	Ah_Kh	8	0.659	200/400	164600	2	8300	3s_Qs_9c_As_Jh_X	ch	3	0	0

Table of 5 rows of observation, mainly focused on player 1 and their information preflop/flop to save space

Preflop Variables (pf)

Player Variables (Variables that repeat for both player 1 and 2):

p_name- Last name of player.

p_stacksize- The size of the player's stack before they bet.

p_position- Player's position on the poker table (how far away they are from the dealer chip), such as bigblind, smallblind, dealer, etc. We used initials like BB or SB to write down the position. We used the chart below to help us determine the position.



p_cards- The 2 cards the player got, written in the format of Card1suite_Card2suite, using initials for the suite (EX: a spade is s) and using J Q K A for Jack, Queen, King, Ace. EX: Jack and 10 of diamonds would be Jd_10d

p_strength- This is the percentage that the player would win against the other player, using perfect information as it knows the cards of both players; these odds are generated by the WSOP organizers. This data was gathered by noting down the % next to the player card. EX: 48% in the picture below.



Nonplayer Variables:

blinds- List the current size of the blinds, in the format of SB/BB. EX: 1500/3000

chip_leader- The number of chips the player with the highest amount of chips at the table has.

last_aggressor- Lists the last player that raised preflop; we put 0 if there were no raises and everyone called the blinds. EX: p1 raised to 1000 chips, then p2 3-bets to 3000 chips, and p1 calls. We would put down 2 for this.

bet_times- How many times a bet was placed preflop. Was this a call, raise, 3-bet, or 4-bet? EX: We would put down 1 if everyone just calls the blinds, 2 if one person raises, 3 if a raise was reraised, etc.

final_bet_amount- This is the size of the final bet preflop. If it were just everyone calling, we would put down a bet size equal to the blind.

pot_size- Size of pot (total amount of chips bet) preflop, after all the betting has completed.

Postflop Variables (Repeated Across Flop[f], Turn[t], River[r])

Player Variables:

p_stack_size- The size of the player's stack before they bet.

p_actions- A list of the player's actions, separated by a "-". We used the abbreviations: ch- check, ca- call, r- raise, f- fold (all in counts as a raise). EX: p1 checks, p2 raises, then p1 reraises and p2 calls. The actions for p1 would be ch-r, and p2 would be r-ca.

p_time- A list of the number of seconds (closest integer) it takes for the player to act, separated by "-". We obtained this by counting the time (using a stopwatch) it took between the cards being shown and p1's action. After p1 makes an action, we would start counting the time it takes p2 to make an action. If p2 bets, we would start counting the time p1 takes to act again. Because these

were manually collected, expect a small margin of error due to human reaction time. EX: Using the *p_actions* example above, if it took 15 seconds to check after the card is shown and 10 seconds to reraise after p2 bets, p1 would have 15-10.

p_strength- Same variable as it was preflop, but updated to take into consideration the community cards revealed.

Nonplayer Variables:

community_cards- Written in the same format as the player's cards, but with 5 cards, where we would put X for any cards that don't get shown. EX: flop comes 1, 2, 3 of spades, and p1 folds. 1s_2s_3s_X_X

ahead- Lists which player is currently winning (using who has the highest *p_strength*), and writes if currently a draw, value would be 0.

pot_size- Size of the pot for the round after all the betting has completed.

Post-hand Variables

winner- Lists down who won the pot, either by the opponent folding or winning at showdown. If it were a draw/chop after revealing the cards, the value would be 0.

final_pot_size- Size of the pot after the hand is over.

Generated Variables

During analysis, we acknowledge that the strength of a hand that was shown on the stream was information that the players would not have access to, as they would not have known their opponents' cards. However, players would know the relative strength of their hand, and to calculate this, we wrote code that computes poker hand strength (HS), defined as the probability that a player's hand is ahead of a random opponent at showdown, using the formula from the University of Alberta Computer Poker Research Group:

$$HS = (ahead + 0.5 * tied) / (ahead + behind + tied)$$

For each observation, the card strings are parsed, and known cards are removed from a standard 52-card deck. We then used a Monte Carlo approach, repeatedly sampling opponent hole cards and any unknown community cards to simulate complete game outcomes. Each simulated hand is evaluated using a full poker hand evaluator that considers all 5-card

combinations and applies standard rankings with kicker-based tie-breaking. Results across simulations were then aggregated to estimate HS at the preflop, flop, turn, and river stages (and blank if it never reached that position), giving us 8 new variables (*pf_p1_hs*, *f_p1_hs*, *t_p1_hs*, *r_p1_hs*, *pf_p2_hs*, *f_p2_hs*, *t_p2_hs*, *r_p2_hs*). We only set the number of Monte Carlo simulations per hand to 500 to save time, leading to roughly a 5% margin of error when compared to other similar online calculators.

Section 2: Summary of the Data

We engineered two bet-sizing metrics to standardize betting behavior across hands with different blind levels and pot sizes. Bet size in big blinds (Bet Size BB = Bet Amount / Big Blind) converts each raw bet amount into the number of big blinds it represents. This makes bet sizes comparable across tournament stages, since a 30,000-chip bet means something very different at 1,000/2,000 blinds than it does at 10,000/20,000 blinds. Bet-size-to-pot ratio (Bet-to-Pot = Bet Amount / Pot Size) measures the bet as a share of the current pot. This captures the pressure a bet puts on an opponent: a half-pot bet (0.5), pot-sized bet (1.0), or overbet (>1.0) can imply very different strategic intentions regardless of the absolute chip amount. Together, these variables allow us to analyze whether players are making small, medium, or large bets relative to both tournament structure and the immediate hand context, which connects directly to our broader goal of studying how stack size, pot size, and bet size influence reaction time and opponent decisions.

Numeric Variables

Variable	N	Minimum	Maximum	Mean	Std. dev.
Bet size faced (BB)	280	0.009	191.667	9.710	16.743
Call price / stack	280	0.000	3.453	0.124	0.261
Prior mean decision time (sec)	280	0.000	59.000	4.519	7.848

Categorical Variables

Variable	Value	Count	Relative frequency
Decision action	fold	134	21.6%
Decision action	raise	332	53.5%
Decision action	call	154	24.8%

Table 1: Summary Statistics for Key Variables

Player Aggression

We first began by exploring whether a player's position in the overall table impacts their playing strategy during a hand. We specifically focused on player aggression, which is the player's tendency to raise bets. We defined a new variable *'gap_diff'* as follows:

$$'gap_diff' = ('p1_stacksize' / 'chip_leader') - ('p2_stacksize' / 'chip_leader')$$

Essentially, this variable compares how far the two players are from the lead in relation to each other. We calculated the *'gap_diff'* for each street: pre-flop, flop, turn, and river. In the pre-flop phase, we used multinomial logistic regression (MNLogit) to regress *'pf_gap_diff'* (pre-flop) with *'last_aggressor'*. When compared to a null model, this resulted in a p-value of 0.37, indicating it was not significant (at a $p < 0.05$ level). For the subsequent streets, we defined another categorical variable called *'raise'* to indicate which player raised during that street, with possible values of 0 (neither), 1, 2, and 3 (both). This variable was defined using the player actions listed for each street. Using the same multinomial logistic regression as before, we then regressed *'f_gap_diff'* with *'f_raise'* for flop, *'t_gap_diff'* with *'t_raise'* for turn, and *'r_gap_diff'* with *'r_raise'* for river. These resulted in p-values of 0.46, 0.87, and 0.68, respectively, indicating that relative stack position alone does not significantly impact player aggression.

For each multinomial regression model, we then added player aggression variables from the previous street. For example, for flop, we regressed *'f_raise'* with both *'f_gap_diff'* and *'last_aggressor'* (from pre-flop). Similarly, *'r_raise'* was regressed with *'r_gap_diff'*, *'t_raise'*, *'f_raise'*, and *'last_aggressor'*. This resulted in models with significant p-values compared to null models for flop ($p=6.47e-7$), turn ($p=0.03$), and river ($p=0.004$). With negative coefficients for a *'raise'=1* response (Player 1 is more likely to raise if they have raised before) and positive coefficients for a *'raise'=2* response (Player 2 is more likely to raise if they have raised before), we see that aggression in a previous street significantly corresponds to aggression in a current street.

We then examined whether this player's aggression was related to the player's perceived hand strength (PHS). This is the strength of the hand as seen by the player themselves, with no information about the other player's hand. We added the PHS variables for each player into the multinomial regression model for each respective street. In pre-flop, we found that PHS was a significant explanatory variable for Player 2's probability of raising ($p=0.002$) but not for Player 1 ($p=0.09$). In flop, we found that PHS was a significant variable for Player 1 ($p=0.03$), raising, but not for Player 2 ($p=0.31$). PHS was not a significant variable for either player during the turn and the river.

Call Frequency

We were interested in how often professional poker players called their opponent off and the weight that they placed on the sizing of the bet. Our figure analyzes call frequency by grouping each faced bet into big-blind buckets and separating the decisions by street: flop, turn, and river. Measuring bet size in big blinds lets us compare hands across different blind levels in the tournament. We observed an overall call rate of 53.9% (151 calls out of 280 call/fold decision spots). The flop had an overall call rate of 57.6%, turn call rate of 53.1%, and river call

rate of 45.5%. We noticed a surprising pattern where the call rates peaked at 5-10 BB, whereas you would assume that the smaller bets with less risk to call would have the highest rates. We concluded that these smaller sizes are viewed as stronger due to the smaller sizing often used to extract value from marginal hands or to induce a re-raise when you have a very strong hand. These smaller sizes are often used on dry boards as well, so the players are less incentivized to call when these small sizes are utilized. With sizes larger than 10 Big Blinds, we saw the expected pattern of call rate decreasing as size increases.

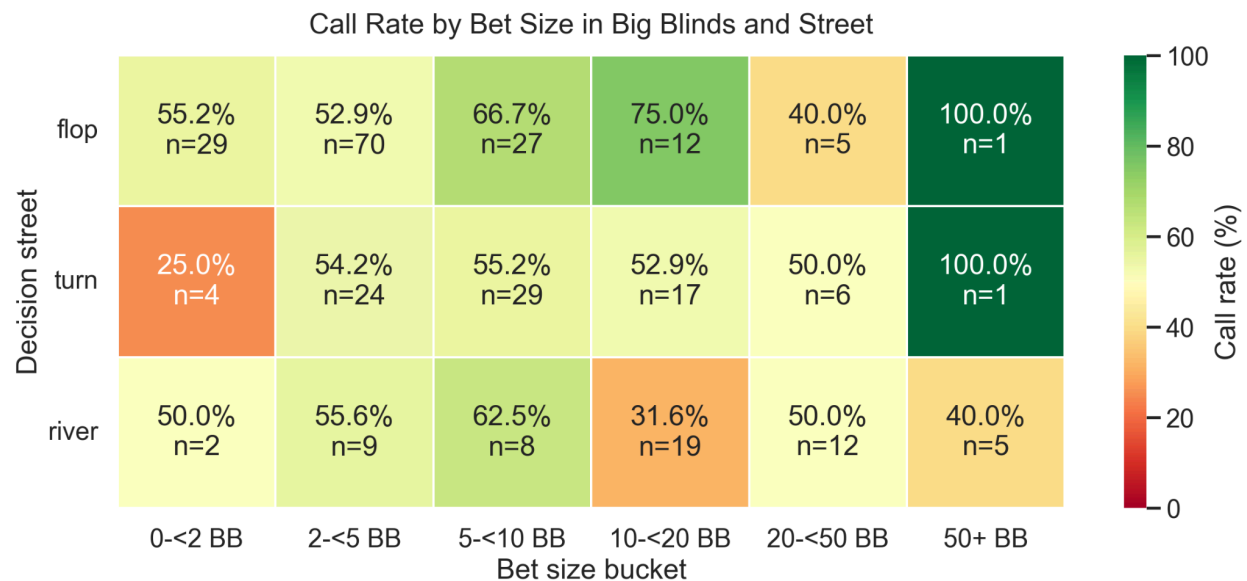


Figure 1: Call rate grouped by bet size and decision street

Play Accuracy

We explored whether players make the “correct” decisions or not based on their true hand strength (THS). While PHS is solely based on the cards that an individual player can see (their own + any community cards), THS compares the cards held by both players and any community cards to determine the strengths of players’ hands relative to each other. This value is shown to the viewer in the livestream, allowing the viewer to see which player objectively has the stronger hand and, as a result, the better chance of winning. We compared the THS of each player during each street to the final resulting winner at the end of the play. This way, if the player with the higher THS wins, we can assume they made the “correct” decisions when playing. On the other hand, if the player with the higher THS loses, we can assume they made “incorrect” decisions when playing. To do this, we define a new variable, ‘*strength_diff*’, which is the difference in THS between Player 1 and Player 2 (as seen below).

$$\text{'strength_diff'} = \text{'p1_strength'} - \text{'p2_strength'}$$

This means that a larger positive number corresponds to a stronger hand for Player 1, while a more negative number corresponds to a stronger hand for Player 2. Since a player's THS changes with each community card reveal, we calculated the difference between the 2 players' THS for each street: pre-flop, flop, turn, and river. The distributions of the THS difference for each street and the corresponding winning player are shown in the figure below. We note that for the river, since all possible cards are revealed, each player's THS is either 0% or 100%, leading to the distribution shown below.

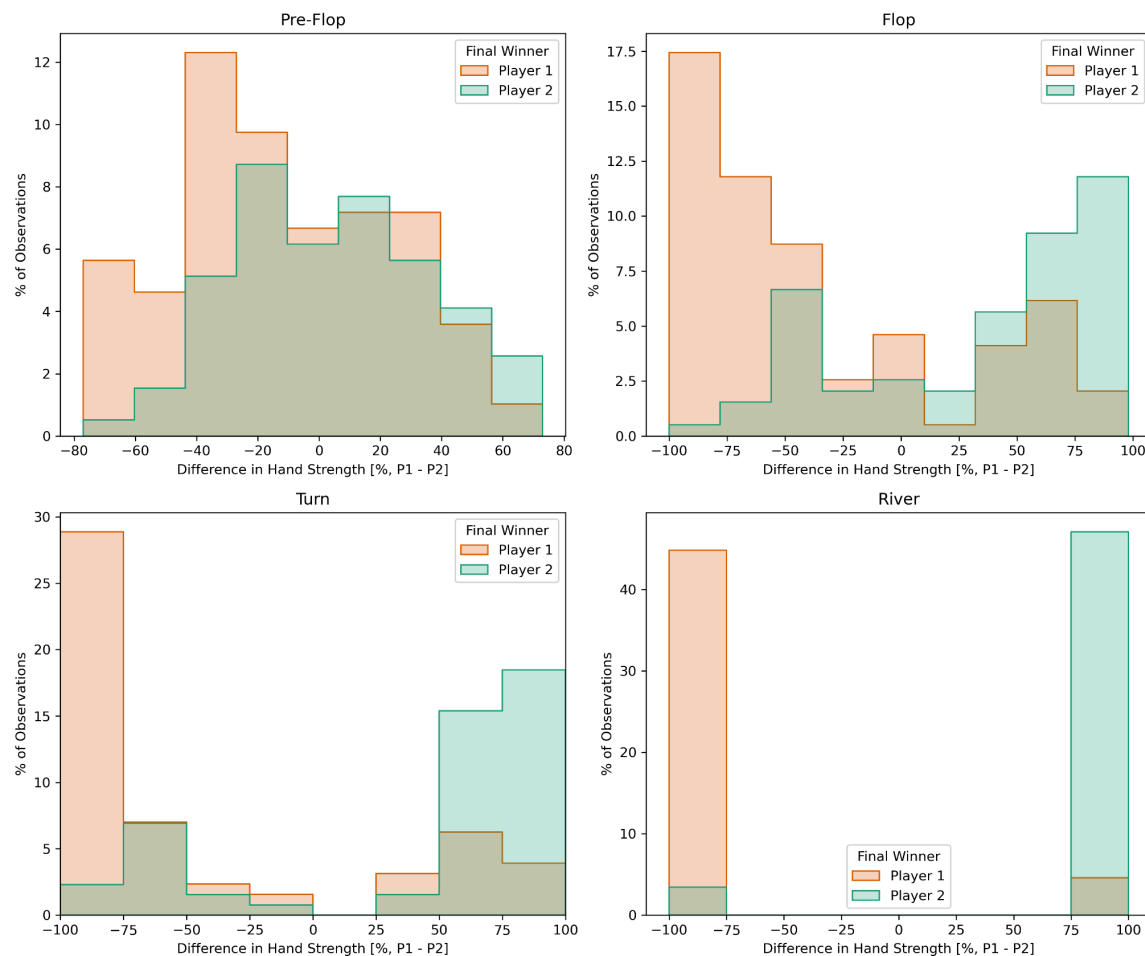


Figure 2: Decision Accuracy by street

Call Rate Prediction

We then wanted to analyze whether our data could help a player predict whether their opponent would call or fold (call = 1). We limited the model to information that would be available to the player and selected prior mean decision time, call price relative to remaining stack, bet-to-pot ratio, and street number. When testing several predictors related to hand strength, decision-making time, bet sizing, player position, and street number, none were statistically significant at the conventional $P < 0.05$ threshold. We were surprised that bet sizing

wasn't a statistically significant variable, as you would expect larger sizes to generate a meaningful increase in opponent folds. After looking at the table of call rates by sizing, we concluded that the parabolic nature of the call rate caused it to overall be a weak predictor, as it wasn't a clear upward linear relationship depending on sizing. It made sense that none of these variables were significant, as we were analyzing skilled players who know how to avoid timing tells and try to stay balanced in their decision-making in order to avoid being exploitable. We kept the model with four features: prior mean decision time, call price relative to remaining stack, bet-to-pot ratio, and street number to balance interpretability and accuracy.

Feature	Coefficient	p-value	95% confidence interval
Prior mean decision time	-0.166	0.278	[-0.465, 0.134]
Call price / stack	-0.071	0.676	[-0.406, 0.263]
Bet size / pot	+0.044	0.794	[-0.285, 0.373]
Street number	-0.010	0.947	[-0.297, 0.277]

Table 2: Coefficients and confidence intervals for logistic regression call rate model

Our modeling attempts really struggled to find much predictive power, and this run had accuracy statistics of F1 Score (overall balance): 0.680, Precision (call correctness): 0.554, Recall (calls found): 0.881, and Accuracy (total correct): 0.554. Predictions were made by grouping any $P > .5$ to call and less than that to fold. The model had a log loss of 0.682, which was also not much better than just guessing a call every time (0.690).

Section 3: Insights from the Data

The following two insights are grounded in the logistic regression models and summary visualizations detailed in Sections 1 and 2. Both insights speak directly to the central purpose outlined in Section 1: identifying actionable, data-backed patterns in WSOP Main Event play that could give a competitive player a meaningful strategic edge.

Insight 1: Prior Mean Decision Time as a Predictor of Opponent Calling Behavior

As shown in *Table 2* from Section 2, a logistic regression model was constructed to predict whether an opponent would call a bet ($\text{Call} = 1$) on a given street, using four features: prior mean decision time, call price relative to remaining stack, bet-to-pot ratio, and street number. Although none of the four features reached conventional statistical significance (all p -values > 0.25), the most practically meaningful result concerns `prior_mean_decision_time`, which carries a negative coefficient of -0.166 (95% CI: $[-0.465, 0.134]$, $p = 0.278$). The direction of this coefficient suggests that opponents who have taken longer to make decisions on earlier streets are less likely to call when they face a bet on a subsequent street, in other words, slower prior decision-making is associated with a tendency to fold rather than continue. While the 95% confidence interval crosses zero, ruling out a definitive claim of significance, the interval extends as low as -0.465 on the lower end, and with a larger sample size, this relationship could plausibly yield a fully negative confidence interval and reach significance. This is directionally consistent with the intuition behind timing tells: a player who deliberates at length is likely wrestling with uncertainty, and that uncertainty ultimately resolves toward folding.

This finding carries a concrete strategic implication. When facing an opponent who has historically taken longer than average to act, a player can increase their bluffing frequency and sizing on later streets, as the model suggests such opponents are more likely to fold. Conversely, when holding a strong hand against the same slow-deciding opponent, a player should consider sizing down their bet to make the call price appear less threatening and to coax a call, maximizing the value extracted. The remaining three features, `call_price_to_stack` ($\beta = -0.071$, $p = 0.676$), `bet_to_pot` ($\beta = 0.044$, $p = 0.794$), and `Street_#` ($\beta = -0.010$, $p = 0.947$), showed no meaningful predictive power in this model. The near-zero, slightly positive coefficient for bet-to-pot is worth noting: conventional poker theory predicts that larger bets relative to the pot should decrease call frequency, yet this was not supported here. This could reflect sample size limitations, but it is also consistent with Figure 1 from Section 2, which shows that call rates do not decrease cleanly as bet sizes in big blinds increase, suggesting that professional players at this level are calibrating their calls on factors beyond raw bet size alone.

Insight 2: Play Accuracy by Street, Where Even Professionals Leave Room for Improvement

Figure 2 in Section 2 presents a four-panel histogram showing the distribution of true hand strength difference ($\text{strength_diff} = \text{P1 Strength} - \text{P2 Strength}$) at each street, separated by which player ultimately won the hand. This figure reveals a striking and consistent pattern: as the hand progresses from pre-flop to river, the final winner aligns more and more closely with the player who held the objectively stronger hand at that street. On the river, where all community cards have been revealed, and each player's THS collapses to either 0% or 100%, the distributions are almost perfectly separated, the orange (Player 1 wins) mass sits squarely at +100, and the teal (Player 2 wins) mass at -100. This is expected, as by the river, there is no ambiguity about who holds the better hand. What is more informative, however, are the pre-flop and flop panels. In the pre-flop panel, the two distributions overlap substantially: Player 1 frequently wins hands even when their pre-flop THS difference is negative (i.e., they were objectively behind), and vice versa. This overlap narrows at the flop and narrows further at the turn, indicating that players' decisions become increasingly aligned with their true hand strength as more information becomes available.

The practical implication of this progression is twofold. First, it suggests that the pre-flop and early-street phases of a hand are where professional players are most susceptible to being outplayed, and where their own decisions are most divergent from pure hand-strength logic. A player who studies pre-flop ranges, continuation betting frequencies, and flop textures stands to gain the most, since this is the window in which deception and strategic noise have the largest influence on outcomes. Second, this figure provides data-based support for the claim that bluffing is most likely to succeed pre-flop and on the flop: because the final winner at these stages is not well-predicted by who currently holds the stronger hand, there is genuine room for a well-timed bluff to succeed even against a strong opponent. By the turn and river, the distributions separate sharply, meaning that the player who is objectively ahead tends to win, suggesting that late-street bluffs against professional players are substantially less likely to succeed. This insight connects to the aggressor analysis in Section 2 as well: since prior aggression significantly predicts future aggression (flop: $p = 6.47 \times 10^{-7}$, turn: $p = 0.03$, river: $p = 0.004$), a player who establishes aggression early, particularly pre-flop and on the flop when outcomes are noisiest, can carry that narrative momentum into later streets with greater credibility, reinforcing the case for early-street initiative.

Critique: What Our Group Would Have Done Differently

Reflecting on the study design and execution, several improvements stand out. Most critically, the feature variables used in the logistic regression for Insight 1 were not standardized before modeling. Because `prior_mean_decision_time` is measured in seconds, `call_price_to_stack` as a proportion, and `bet_to_pot` as a ratio, the raw coefficients are not directly comparable in scale. Standardizing all features to a mean of zero and a standard

deviation of one would have made the coefficients interpretable on the same scale. Second, our effective sample size for the logistic regression was considerably smaller than 200, since the model required observations where a bet occurred and a call-or-fold decision was recorded. Third, our definition of a “correct” decision in Insight 2 was limited because it did not account for bluffs. We primarily judged correctness using true hand strength (THS), but poker strategy is not always based on having the objectively stronger hand. A player may be behind in THS and still make the strategically correct decision by bluffing if they believe their opponent is weak or likely to fold. Future models could account for this by incorporating perceived hand strength (PHS) into the definition of correct play. For example, if both players’ PHS values are around or below 0.5, neither player appears to have a clearly strong hand from their own perspective, so a bluff could be considered a reasonable or even correct move regardless of THS. Finally, for Insight 2, formally testing whether correct decision rates differ significantly across streets, through a proportion test or chi-square test, would have elevated Figure 3 from a visually compelling observation to a statistically defensible claim.

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